

Calibrating Learners' Trust in Generative AI: A Narrative Review of Design Interventions for Appropriate Reliance in Education

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Abstract

Background. Students now use generative AI (GenAI) to explain concepts, check answers, write code, and draft essays. Whether this helps or harms depends less on how much students trust these systems than on whether their trust is *calibrated*: relying on AI when it is right, questioning it when it is wrong, and declining to use it when doing the work oneself is the point of the exercise. Human-computer interaction (HCI) research has developed many interventions to foster appropriate reliance on AI, but most were tested on decision-making or question-answering tasks rather than on learning tasks.

Aim. In this narrative review, I ask which design interventions calibrate learners' trust in and reliance on GenAI, what outcomes they have been shown to change, and whether their benefits are shared equally across learners.

Approach. Literature was identified through Crossref, OpenAlex, PubMed, and arXiv searches, supplemented by targeted searches for frequently cited studies on trust calibration and generative AI in education. Only peer-reviewed sources are cited. The review is narrative and non-systematic; its search is described so readers can judge coverage.

Findings. Evidence from high school and university settings shows poorly calibrated reliance: among 432 undergraduates, higher trust was associated with worse discrimination between correct and incorrect AI suggestions, and unrestricted AI access during practice harmed later performance in a field experiment with high school students. Generative AI adds its own difficulties: fluent and lengthy answers inflate confidence, models rarely express uncertainty and are overconfident when they do, and trust recovers slowly after confident errors. Interventions fall into four families: communicating reliability (uncertainty expressions, calibrated explanations, sources), forcing cognitive engagement (cognitive forcing, questioning, guarded tutors, metacognitive prompts), changing the costs and benefits of verification, and building learner capacity (AI literacy). Explanations alone tend to increase reliance whether the AI is right or wrong. Interventions that make users think before accepting an answer reduce overreliance more reliably, but users rate them less favorably, and several studies find they help learners with higher need for cognition, prior knowledge, or AI literacy more. AI literacy moderates the link between trust and appropriate reliance, but few AI literacy programs assess what students actually learned.

Conclusions. I propose that appropriate reliance in education has two reference points, the AI's reliability and the learner's learning goal (summarized in Figure 1), and that most interventions address only the first. Progress requires outcome measures that separate trust level from calibration and from learning, evaluation with AI access removed, and routine checks for intervention-generated inequalities.

Keywords: trust calibration; appropriate reliance; overreliance; generative AI; education; metacognition; human-AI interaction; equity

1. Introduction

Trust has long been treated as the variable that governs how people use automation. Classic work distinguished *misuse*, overreliance that leads to failures of monitoring, from *disuse*, the neglect or underutilization of automation (Parasuraman & Riley, 1997). Lee and See (2004) argued that because complete understanding of an automated system is often impractical, trust guides reliance, and they framed the design problem as one of *appropriate* reliance rather than more reliance. Later syntheses organized the many factors that shape trust into dispositional, situational, and learned layers (Hoff & Bashir, 2015) and distinguished warranted trust, grounded in a system's actual trustworthiness, from unwarranted trust (Jacovi et al., 2021).

Generative AI has moved this question into classrooms. Students can now ask a large language model (LLM) to solve a problem, explain an error, or summarize a reading, and the model will answer fluently whether or not it is right. Evidence on the educational value of these tools is genuinely mixed. A randomized trial with college students found that they learned more in less time with a carefully designed AI tutor than in an active-learning class (Kestin et al., 2025), and ChatGPT-generated hints in mathematics produced learning gains with no statistically significant difference from human-authored hints, even though ChatGPT-generated help initially failed quality checks on 32% of problems (Pardos & Bhandari, 2024). Yet a field experiment with nearly a thousand high school students found that unrestricted access to GPT-4 during practice improved performance while access lasted but left students performing worse than controls once access was removed (Bastani et al., 2025). Students themselves are ambivalent. In a survey of 399 university students in Hong Kong, attitudes toward GenAI were generally positive, but students voiced concerns about accuracy, privacy, and the effect on their personal development (Chan & Hu, 2023).

These contrasting results point to the same underlying issue. The benefit of AI in learning depends on *how* students rely on it, and how they rely on it depends on trust that may or may not be warranted. Several recent reviews address parts of this problem. Mehrotra et al. (2024) systematically reviewed approaches for fostering appropriate trust in human-AI interaction, including confidence scores, explanations, and uncertainty communication. Wischniewski et al. (2023) reviewed 96 studies of trust calibration interventions for automated systems and concluded that how calibration is measured often limits what can be concluded about interventions. In education, Zhai et al. (2024) reviewed 14 studies on how overreliance on AI dialogue systems affects students' cognitive abilities. What is missing is a synthesis that asks, specifically for learners using GenAI, which interventions have been tested, what they actually change (trust level, the appropriateness of reliance, or learning itself), and for whom they work.

I address that gap with three aims:

1. To clarify what “appropriate reliance” should mean when the user is a learner, drawing on the human-automation trust tradition and recent work on GenAI and metacognition.
2. To synthesize evidence on design interventions intended to calibrate trust and reliance, distinguishing evidence from learning contexts from evidence gathered in general decision tasks.
3. To examine whether intervention effects differ across learners, and to propose a research agenda that treats equitable calibration as an explicit design goal.

2. Approach to the Literature

This is a narrative review: my goal is to interpret and connect a body of work, not to estimate pooled effects. Following guidance that narrative reviews report how literature was identified (Baethge et al., 2019), the search is summarized here.

Sources and dates. Searches were run in September 2026 in Crossref, OpenAlex, and PubMed, and arXiv was searched to identify peer-reviewed versions of recent work. Search terms combined concepts of *trust and reliance* (“trust,” “trust calibration,” “appropriate reliance,” “overreliance”), *AI* (“generative AI,” “large language model,” “ChatGPT,” “AI assistant”), *learning* (“students,” “education,” “learning,” “novice programmers”), and *intervention* (“explanation,” “uncertainty,” “cognitive forcing,” “metacognitive prompts,” “AI literacy”). These were supplemented by targeted title searches for frequently cited studies in the trust-calibration and AI-in-education literatures.

Selection. Studies were included if they (a) measured trust in or reliance on AI among learners, (b) tested an intervention intended to change reliance on AI or engagement with AI output, in any population, where the mechanism was plausibly relevant to learning, or (c) measured learning or skill outcomes after AI-assisted work. Conceptual papers were included where they defined terms used across the literature.

Limitations of the approach. The search was not exhaustive and was limited to English-language sources. Preprints and working papers were excluded, so some recent findings not yet peer reviewed are not represented. Findings are reported as stated by the original authors; where I draw my own inferences, the text says so.

3. What Counts as Appropriate Reliance for a Learner?

3.1 Trust, reliance, and calibration

Three terms need to be kept apart. *Trust* is an attitude: a belief or expectation about a system. *Reliance* is a behavior: accepting or acting on the system's output. *Calibration* describes the fit between the two and the system's actual capabilities: to achieve appropriate reliance, users' trust should reflect what the system can do (Wischnewski et al., 2023). The distinction matters because many studies measure only one. A survey of empirical methods for studying trust in AI-assisted decision-making found that definitions of trust were often not integrated into experimental protocols, making findings hard to compare (Vereschak et al., 2021). Schemmer et al. (2023) proposed Appropriateness of Reliance as a quantifiable, two-dimensional measurement concept, reflecting that people must both rely on correct AI advice and turn down incorrect advice. Mehrotra et al. (2024) similarly found no single agreed definition of appropriate trust and proposed mapping related concepts across beliefs, intentions, and actions.

For GenAI, a further distinction is useful. Steyvers et al. (2025) separate the *calibration gap*, the difference between people's confidence in an LLM's answers and the model's own confidence, from the *discrimination gap*, how well people can tell correct from incorrect answers. A learner can be well calibrated on average while still failing to discriminate between individual right and wrong answers, and these call for different remedies.

3.2 Why learning adds a second reference point

In decision-support research, reliance is usually judged against a single standard: was the AI correct? Learning complicates this. The purpose of a practice problem is not to produce a correct answer but to build the capacity to produce correct answers later. Cognitive offloading is defined

as “the use of physical action to alter the information processing requirements of a task so as to reduce cognitive demand” (Risko & Gilbert, 2016). In a learning task, however, the cognitive demand is frequently the thing being practiced.

I therefore propose that appropriate reliance for learners has two reference points. The first is the **reliability of the AI**: accepting correct output and rejecting incorrect output. The second is the **learning goal**: relying on AI for components of a task that are not the learning target, and doing the targeted work oneself. A student who accepts a correct AI solution to a practice problem is appropriately reliant by the first standard and inappropriately reliant by the second. Bastani et al. (2025) illustrate the distinction: students with unrestricted GPT-4 access performed better with the tool but worse afterward, using it as what the authors call a “crutch.”

This two-reference-point framing is my own synthesis, not an established construct. I offer it because most of the interventions reviewed below were designed to address only the first reference point. Figure 1 summarizes the four resulting situations.

Figure 1. Two reference points for appropriate reliance in learning (proposed here).

AI output	Component is not the learning target	Component is the learning target
Correct	Rely on the AI. Appropriate by both standards.	Attempt first, then compare with the AI. Relying would be accurate but would skip the practice.
Incorrect	Verify and reject. Requires the learner to detect the error.	Attempt first, then detect and reject. The error becomes a learning opportunity only if it is caught.

Note. Decision-support research evaluates reliance only by rows (was the AI correct?). Learning adds the columns. In my reading, the learning losses from unrestricted AI use during practice reported by Bastani et al. (2025) arise largely in the right-hand column.

3.3 Why generative AI makes calibration harder

Several properties of current LLM-based systems make calibrated trust harder to achieve than with earlier forms of automation.

Fluency without calibrated confidence. Users overestimate the accuracy of LLM answers, and longer explanations raise confidence even when they add no accuracy (Steyvers et al., 2025). Models themselves rarely signal doubt. An analysis of publicly deployed language models found that they were reluctant to express uncertainty even when their answers were wrong; when prompted to state their confidence they tended to be overconfident, with an average error rate of 47% among confident responses (Zhou et al., 2024). In the same work, users relied heavily on model outputs whether or not certainty was marked, and the human preference data used to fine-tune models showed a bias against text expressing uncertainty. The signal that calibration depends on is therefore often missing at the source.

Beliefs and affect shape trust independently of reliability. Trust in AI has cognitive and emotional components. A review of empirical research linked tangibility, transparency, reliability, and immediacy behaviors to cognitive trust, and anthropomorphism specifically to emotional trust (Glikson & Woolley, 2020). How a system is introduced also matters. In one experiment, participants interacted with the same conversational AI after being primed with statements that it had caring, manipulative, or no motives; those who perceived a caring motive also perceived the AI as more trustworthy, empathetic, and better-performing, and the effects of priming were stronger for a more sophisticated model (Pataranutaporn et al., 2023). In my reading, tutors designed to feel warm and supportive may therefore raise trust in ways unrelated to their accuracy.

Trust is dynamic, and errors cost more than they should. In a betting game, a few incorrect predictions made with inaccurate confidence damaged user trust and performance, and recovery was very slow; confidently incorrect and unconfidently correct outputs harmed trust in different ways (Dhuliawala et al., 2023). Over a semester, a student's trust is shaped by a history of encounters with an AI's errors, which single-session studies cannot capture.

4. How Well Calibrated Is Learners' Reliance?

Trust predicts worse discrimination. In a study of 432 undergraduates working on Python output-prediction problems with an AI assistant that gave both accurate and deliberately misleading suggestions, higher trust was associated with *lower* appropriate reliance, meaning weaker discrimination between correct and incorrect suggestions (Pitts et al., 2026b). An earlier study by the same group found that appropriate reliance was related to programming self-efficacy, programming literacy, and need for cognition, while overreliance correlated with post-task trust and satisfaction (Pitts et al., 2026a). Among university computer science students, survey research found varying levels of trust in GenAI, alongside different levels of confidence and motivation (Amoozadeh et al., 2024).

Fluency inflates confidence. In experiments with LLM answers to multiple-choice and short-answer questions, users overestimated the accuracy of responses when given default explanations, and longer explanations increased confidence even when the extra length did not improve accuracy (Steyvers et al., 2025). In fact-checking, users reading LLM explanations were more efficient than those using a search engine but overrelied on the LLM when its explanation was wrong (Si et al., 2024).

Reliance can create an illusion of competence. In observational lab sessions with novice programmers, students who were not struggling used GenAI to accelerate work they already understood and ignored unhelpful suggestions, while struggling students experienced compounded metacognitive difficulties, overestimated their performance, and finished with what the authors describe as an illusion of competence (Prather et al., 2024). A survey of 319 knowledge workers found that higher confidence in GenAI was associated with less critical thinking, while higher self-confidence was associated with more (Lee et al., 2025). Although that sample was not students, the pattern is consistent with the student findings.

Reliance can displace learning. In a randomized experiment with 1,625 students across 10 courses, students tended to rely on AI prompts for writing peer feedback rather than learn from them; when the AI was removed, self-monitoring checklists partly filled the gap but were less effective than the AI assistance itself (Darvishi et al., 2024). University students using ChatGPT for a writing task improved their essay scores more than comparison groups, but their knowledge gain and transfer did not differ, and the authors warn of “metacognitive laziness” (Fan et al., 2024). Students researching a socio-scientific issue with ChatGPT experienced lower cognitive load than those using a search engine but produced lower-quality reasoning (Stadler et al., 2024). Survey evidence links frequent AI tool use with lower critical thinking scores, mediated by cognitive offloading (Gerlich, 2025); because that evidence is correlational, it cannot show that AI use caused the difference.

But well-designed AI can support learning. The same literature contains strong positive results when tools are designed around pedagogy: the AI tutor designed around the same pedagogical practices as in-class active learning (Kestin et al., 2025), and the guarded “GPT Tutor” condition that largely mitigated the negative learning effects of unrestricted access (Bastani et al., 2025).

The question is therefore not whether students should use GenAI but which designs calibrate their reliance.

5. Interventions: What Has Been Tried and What It Changes

The interventions in this literature can be grouped into four families by mechanism. Table 1 summarizes representative studies, noting whether each was tested with learners and which outcome it changed.

Table 1. Representative interventions for calibrating reliance on AI, by mechanism.

Mechanism	Intervention (study)	Population and task	Main reported effect
Communicate reliability	First-person uncertainty expressions (Kim et al., 2024)	Medical questions with LLM-infused search (N = 404)	Lower agreement with the system and higher accuracy; reduced overreliance
Communicate reliability	Explanations adjusted to model confidence (Steyvers et al., 2025)	Multiple-choice and short-answer questions	Narrowed calibration and discrimination gaps
Communicate reliability	Sources; inconsistent explanations (Kim et al., 2025)	Objective questions with an LLM-infused application (N = 308)	Less reliance on incorrect answers; explanations alone increased reliance on both correct and incorrect answers
Force engagement	Cognitive forcing functions (Buçinca et al., 2021)	AI-assisted decision task (N = 199)	Reduced overreliance vs. simple explanations; lowest subjective ratings; benefited high need-for-cognition users more
Force engagement	AI-framed questioning (Danry et al., 2023)	Judging logical validity of statements (N = 204)	Improved discernment of flawed statements vs. causal explanations
Force engagement	Guarded GPT tutor (Bastani et al., 2025)	High school math students	Largely mitigated learning losses after access was removed
Force engagement	Step-by-step guided engagement with AI code (Kazemitabaar et al., 2025)	Novice programmers	Identified as the most effective of the techniques tested, evaluated on friction, transfer without AI, and alignment of perceived with actual ability
Force engagement	Attempt before seeing LLM explanation (Kumar	1,200 participants, math	Largest learning benefit when problems were

Mechanism	Intervention (study)	Population and task	Main reported effect
	et al., 2025)	problems	attempted first
Force engagement	Metacognitive prompts in GenAI search (Singh et al., 2025)	University students	More active engagement and deeper inquiry; varied with metacognitive flexibility
Force engagement	Contrastive explanations anticipating misconceptions (Buçinca et al., 2025)	Decision task (N = 628)	Improved independent decision-making skill without lowering accuracy
Build capacity	Self-monitoring checklists (Darvishi et al., 2024)	University students, peer feedback	Partly filled the gap when AI was removed; less effective than AI assistance
Build capacity	Priming beliefs about the AI's motives (Pataranutaporn et al., 2023)	Interaction with a conversational AI	Perceiving a caring motive went with higher perceived trustworthiness, empathy, and performance of the same system

5.1 Communicating reliability

The most intuitive family of interventions tells users how reliable the AI is. The evidence shows this can work, but the form matters. In a pre-registered experiment, first-person expressions of uncertainty (“I’m not sure, but...”) reduced agreement with an LLM-infused search engine and increased accuracy, apparently by reducing overreliance on incorrect answers; uncertainty phrased impersonally had weaker, non-significant effects (Kim et al., 2024). Showing confidence scores helped calibrate trust in an earlier decision-support study, but trust calibration alone was not sufficient to improve AI-assisted decisions, which may also depend on whether humans bring knowledge that complements the AI (Zhang et al., 2020). Adjusting LLM explanations to reflect the model’s internal confidence narrowed both calibration and discrimination gaps (Steyvers et al., 2025).

Explanations deserve particular caution. In user studies on three datasets, explanations increased the chance that people accepted AI recommendations regardless of correctness (Bansal et al.,

2021), and in a recent LLM experiment, explanations increased reliance on both correct and incorrect answers (Kim et al., 2025). The same experiment found less reliance on incorrect answers when sources were provided or when explanations contained inconsistencies. Presenting contrastive information, arguments for why a claim is both true and false, mitigated overreliance in fact-checking (Si et al., 2024). In code, highlighting tokens most likely to be edited helped programmers, while highlighting tokens by generation probability gave no benefit (Vasconcelos et al., 2024), a reminder that exposing a model’s raw uncertainty is not the same as communicating useful uncertainty.

Assessment. Reliability communication is the best-studied family, but almost all of this evidence comes from decision-making or question-answering tasks rather than learning tasks. None of the reliability-communication studies in Table 1 measured learning with the AI removed.

5.2 Forcing cognitive engagement

A second family aims to make users think before they accept. Bućinca et al. (2021) argued, drawing on dual-process theory, that people rarely engage analytically with each AI recommendation and instead develop heuristics about when to follow it. Their cognitive forcing designs, which required people to engage with the task before or while seeing AI output, reduced overreliance compared with simple explainable AI approaches. Two findings from that study recur throughout this review: participants gave the least favorable ratings to the designs that reduced overreliance most, and the interventions benefited people with higher need for cognition more.

Related designs reframe the AI’s role. AI-framed questioning, which turns information relevant to an AI judgment into questions for the user, improved people’s discernment of logically flawed

statements compared with causal explanations, even from an always-correct system (Danry et al., 2023). Contrastive explanations that anticipate a user's likely misconception improved people's independent decision-making skills without lowering decision accuracy (Bućinca et al., 2025). This is one of the few studies in the general literature to measure skill development directly.

In education, the evidence for this family is the strongest available. The guarded "GPT Tutor" in Bastani et al. (2025), a randomized trial at a high school in Turkey, largely mitigated the learning losses seen with unrestricted access; its main safeguard was instructing the tutor to give teacher-designed hints instead of answers. A classroom deployment of CodeAid, an LLM programming assistant designed not to reveal code solutions, across 700 students produced design principles including avoiding direct answers while encouraging motivated learning (Kazemitabaar et al., 2024). A follow-up study compared seven cognitive engagement techniques on the friction they introduced, learners' ability to apply concepts to isomorphic tasks without AI, and the alignment of learners' perceived and actual coding ability; guiding learners step by step, prompting them to say what needed to be done at each stage before the corresponding code was revealed, emerged as the most effective technique (Kazemitabaar et al., 2025). Metacognitive prompts inserted into GenAI search led university students to explore more topics and ask deeper follow-up questions, with effects that depended on students' metacognitive flexibility (Singh et al., 2025). The order of work matters. In a pre-registered experiment with 1,200 participants solving math practice problems, LLM-generated explanations improved learning compared with seeing only correct answers, and the largest benefits came when participants attempted problems on their own before consulting the explanations (Kumar et al., 2025). At course scale, Harvard's CS50 deployed course-specific AI tools designed to guide students toward solutions rather than offer them outright, while limiting the use of commercial AI tools; students reported that the tools felt like

having a personal tutor (Liu et al., 2024). That report describes deployment and student perceptions rather than measured learning or calibration outcomes. These designs respond to the argument that GenAI imposes high metacognitive demands on users and that systems should support metacognitive monitoring and control (Tankelevitch et al., 2024).

Assessment. Engagement-forcing designs are the only family with direct evidence of preserved learning, and they address the second reference point, the learning goal, as well as the first. Their costs are friction and lower user satisfaction, which in voluntary settings may lead students to choose unrestricted tools instead.

5.3 Changing the costs and benefits of verification

A third perspective treats overreliance as a rational choice rather than a bias. Across five studies, Vasconcelos et al. (2023) showed that task difficulty, explanation difficulty, and monetary incentives changed overreliance, and argued that people strategically weigh the cost of verifying AI output against the benefit. They argue that explanations fail to reduce overreliance when they do not sufficiently lower the cost of verifying the AI's output. For learners, this suggests that assignments and grading shape reliance as much as interface design: if verification is costly and unrewarded, even a well-calibrated student may rationally rely on the AI. This mechanism has not, to my knowledge, been tested directly with GenAI in a course setting.

5.4 Building learner capacity

The fourth family targets the learner rather than the interface. AI literacy has been defined as a set of competencies that enable people to critically evaluate AI and use it effectively (Long & Magerko, 2020), and reviews have organized it around knowing and understanding, using and applying, evaluating and creating, and ethical issues (Ng et al., 2021). A systematic review of

179 documents on AI literacy in K-12 education found that very few learning experiences assessed whether students actually understood AI concepts afterward (Casal-Otero et al., 2023).

AI literacy does appear to matter as a learner characteristic: it significantly moderated the relationship between trust and appropriate reliance among undergraduates (Pitts et al., 2026b). Whether AI literacy *programs* produce that benefit is a separate question, and the K-12 review above suggests it is rarely tested. Literacy interventions should be evaluated on calibration, not only on knowledge or skepticism, since greater skepticism alone could shift students from overreliance to rejecting correct advice.

A related lever is how an AI system is framed when students first meet it. Because priming beliefs about an AI's motives shaped how the same system was experienced (Pataranutaporn et al., 2023), the introduction of a classroom AI tool, including what the instructor says about its reliability and purpose, is itself a trust intervention. It is rarely treated as one.

5.5 Teachers and institutions as part of the calibration system

Students do not calibrate their trust alone. Teachers decide which tools are allowed, how assignments reward verification, and how AI is described in class. Teachers' own trust has also been studied. In a professional development program for K-12 science teachers working with an AI grading tool, explaining how the AI makes decisions compared with human experts, and how it could complement rather than replace teachers, reduced concerns and improved trust (Nazaretsky et al., 2022). That program aimed to increase trust, not to calibrate it, which illustrates how the goal of adoption can differ from the goal of appropriate reliance. Institutional choices also shape reliance: CS50's policy of encouraging course-specific tools while limiting commercial ones (Liu et al., 2024) is effectively a decision about which system students will

learn to trust. The cost-benefit account of overreliance (Vasconcelos et al., 2023) suggests that grading policies, by changing the payoff for verifying AI output, may matter as much as interface design. None of these institutional levers has been evaluated on students' calibration.

5.6 Across the four families: three patterns

Read together, the intervention studies show three patterns that a single family would not reveal.

Adding information is not the same as adding thought. Interventions that give learners more information about the AI, such as explanations, sources, or confidence, work only if learners engage with that information. Explanations increased reliance regardless of correctness in two separate studies (Bansal et al., 2021; Kim et al., 2025), while designs that required people to think before or while seeing AI output reduced overreliance (Buçinca et al., 2021) or improved learning (Kazemitabaar et al., 2025; Kumar et al., 2025). The cost-benefit account offers a possible explanation: information may help only when it lowers the cost of verification enough to make checking worthwhile (Vasconcelos et al., 2023). In learning, the practical implication is that a reliability cue attached to an answer the student can simply copy may do little, while the same cue attached to a task that requires the student's own attempt may do more.

What works is often what users like least. The designs that reduced overreliance most received the least favorable ratings (Buçinca et al., 2021), and among students, overreliance has been found to correlate with post-task trust and satisfaction (Pitts et al., 2026a). If satisfaction is used to choose tools, it may select for the designs that calibrate trust worst. This tension deserves direct study.

Evidence is strongest where it is least transferable, and weakest where it matters most. The best-controlled evidence on reliability communication comes from general decision and

question-answering tasks, while the most important outcome for education, durable learning measured without AI, appears mainly in a small number of education-specific studies of engagement-forcing designs. Among the studies reviewed here, very few measure both calibration and learning in the same design. Closing this gap is the clearest opportunity in the field.

6. For Whom Do Interventions Work?

A recurring finding across the literature is that the benefits of AI assistance and of calibration interventions are not evenly distributed.

Individual differences moderate intervention effects. Cognitive forcing benefited people higher in need for cognition more (Bućinca et al., 2021). Among students, the relationship between trust and appropriate reliance was moderated by AI literacy and need for cognition (Pitts et al., 2026b), and metacognitive prompts worked differently depending on metacognitive flexibility (Singh et al., 2025).

Prior knowledge shapes who gains. Struggling novice programmers were the ones for whom GenAI compounded metacognitive difficulties, while students who were not struggling used it to accelerate work they already understood (Prather et al., 2024). Trust itself varies by group: in a validated instrument study of 665 students, gender and educational background were significantly correlated with trust in AI-powered educational technology (Nazaretsky et al., 2025).

Inequality may begin before calibration. Who uses GenAI at all is also unequal. In a survey of 18,000 Danish workers in exposed occupations linked to register data, women were 16 percentage points less likely than men to have used ChatGPT for work, and users had already

earned slightly more before ChatGPT arrived; workers cited employer restrictions and a perceived need for training as barriers (Humlum & Vestergaard, 2025). These are workers, not students, but the finding is a reminder that calibration interventions reach only those who use the tools, and that access and training gaps may compound differences in calibration.

Work and learning may diverge. This pattern contrasts with evidence from the workplace, where GenAI has tended to compress performance differences. In a writing experiment with professionals, inequality between workers decreased with ChatGPT access (Noy & Zhang, 2023), and among customer-support agents, less experienced and lower-skilled workers gained the most (Brynjolfsson et al., 2025). A plausible reconciliation, which I offer as a hypothesis rather than a finding, is that AI compresses differences in *output*, which is what workplace studies measure, while it can widen differences in *learning*, because the learners who most need to practice are also the most likely to substitute AI for practice.

Friction may sort learners. If engagement-forcing designs are the most effective but least liked (Buçinca et al., 2021), and if they help learners with higher need for cognition more, then in voluntary settings they risk reaching mainly the learners who need them least. Buçinca et al. (2021) explicitly audited their work for such “intervention-generated inequalities.” Explicit audits of this kind are uncommon among the education studies reviewed here, although moderation analyses such as those of Pitts et al. (2026b) point in the same direction; they should become standard.

Toward equitable calibration. If interventions help unevenly, one response is to adapt them. A system could apply more scaffolding, such as requiring an attempt before revealing a solution, when signals suggest a student is substituting rather than complementing, and less when a student is using AI to check work they already understand. The divide observed by Prather et

al. (2024), between students who used GenAI to accelerate work they understood and students for whom it compounded difficulties, suggests what such signals might look like. Adaptive designs carry their own risks, including misclassifying students and imposing friction unevenly, so they should be evaluated with the same subgroup analyses this review recommends for fixed designs.

7. Measurement Problems in the Evidence Base

Several measurement issues limit what the current literature can conclude.

Trust level is not calibration. Some education studies measure self-reported trust or readiness to adopt AI (e.g., Amoozadeh et al., 2024; Nazaretsky et al., 2025), which is valuable but does not show whether reliance is appropriate. Reviews of trust calibration research found that how calibration is measured often limits interpretation of interventions (Wischnewski et al., 2023), and that definitions of trust are often missing from study protocols (Vereschak et al., 2021). Behavioral paradigms that seed incorrect AI suggestions, as in Pitts et al. (2026a, 2026b), are more informative.

Assisted performance is not learning. Studies that measure only performance with AI present can be misleading, as the contrast between the two phases of Bastani et al. (2025) shows. The most informative designs measure transfer with AI access removed (Bastani et al., 2025; Darvishi et al., 2024; Kazemitabaar et al., 2025).

Short studies dominate. A review of empirical research on trust in AI noted overreliance on short-term, small-sample experiments, where trust develops differently from longer-term, higher-stakes settings (Glikson & Woolley, 2020). Most intervention studies in Table 1 are single-session experiments. Semester-long deployments (Kazemitabaar et al., 2024) and field

experiments (Bastani et al., 2025; Kestin et al., 2025) remain the exception. Because trust erodes quickly after confident errors and recovers slowly (Dhuliawala et al., 2023), single-session studies may misstate how interventions perform once students have experienced an AI’s failures.

Table 2 summarizes the constructs that a study of calibration in learning should distinguish, with an example measure for each.

Table 2. Constructs to measure separately in studies of learners’ reliance on AI.

Construct	What it captures	Example measure	Example study
Trust	Attitude toward the system	Validated questionnaire	Nazaretsky et al. (2025)
Reliance	Behavior	Agreement with AI on seeded correct and incorrect items	Pitts et al. (2026a, 2026b)
Appropriateness of reliance	Whether behavior fits AI correctness	Two-dimensional appropriateness-of-reliance measure	Schemmer et al. (2023)
Calibration and discrimination	Fit between user confidence and AI accuracy	Calibration gap; ability to separate correct from incorrect answers	Steyvers et al. (2025)
Metacognitive monitoring	Accuracy of self-assessment	Gap between perceived and actual ability	Kazemitabaar et al. (2025)
Learning	Durable skill	Transfer test with AI access removed	Bastani et al. (2025); Kazemitabaar et al. (2025)
Distribution of effects	Who benefits	Effects by prior knowledge, AI literacy, need for cognition	Buçinca et al. (2021); Pitts et al. (2026b)

8. Implications for Practice

Although the evidence base is still developing, several implications for educators, designers, and institutions follow from the studies reviewed here. I offer them as reasonable inferences, not as tested guidelines.

For designers of educational AI. Treat explanations with caution, since they can raise reliance on wrong answers as much as on right ones (Bansal et al., 2021; Kim et al., 2025). Prefer designs

that ask for the learner's own thinking before revealing a solution (Kazemitabaar et al., 2025; Kumar et al., 2025), and build in safeguards for learning, as in the guarded tutor of Bastani et al. (2025). Where the system communicates reliability, use first-person uncertainty and point to sources (Kim et al., 2024, 2025), and check whether the model expresses uncertainty at all, since many do not by default (Zhou et al., 2024).

For instructors. Decide, for each assignment, which parts are the learning target, and tell students. This makes the second reference point in Figure 1 explicit rather than leaving students to infer it. Consider how AI tools are introduced, since framing shapes trust (Pataranutaporn et al., 2023), and consider assessment designs that reward verifying AI output (Vasconcelos et al., 2023). Include some AI-free assessment, since assisted performance can hide lost learning (Bastani et al., 2025).

For institutions. Evaluate AI tools on students' calibration and learning, not only on satisfaction or usage, which may favor the designs that calibrate trust least (Buçinca et al., 2021; Pitts et al., 2026a). Pay attention to who is not using the tools, and who uses them without guidance, since both access and benefit appear to be unevenly distributed (Humlum & Vestergaard, 2025; Prather et al., 2024).

9. Future Directions

1. Test decision-support interventions with learners. The best-studied reliability interventions, including uncertainty expressions, calibrated explanations, sources, and contrastive information, have rarely been tested with students on learning tasks. Replications that add delayed, AI-free transfer measures would show whether they calibrate only assisted performance or also support learning.

2. Design for both reference points. Interventions should be judged on whether they help learners accept correct output, reject incorrect output, *and* do the work that is the learning target. Designs that adapt what the AI reveals to the learning goal, such as guarded tutors and step-by-step engagement, are promising templates.

3. Report calibration, not only trust. Studies should seed incorrect AI output, report appropriate reliance on both correct and incorrect cases (Schemmer et al., 2023), and distinguish calibration from discrimination (Steyvers et al., 2025).

4. Audit for intervention-generated inequalities. Every intervention study should report effects separately by prior knowledge, AI literacy, and need for cognition at minimum, following the audit practice of Bućinca et al. (2021). Where effects differ, adaptive designs that apply more scaffolding to learners at risk of substitution should be tested.

5. Evaluate AI literacy on calibration. Few K-12 AI literacy programs assess what students learned (Casal-Otero et al., 2023), yet AI literacy moderates how trust translates into reliance (Pitts et al., 2026b). Programs should be tested on whether they improve appropriate reliance on both correct and incorrect AI output, not only on knowledge or skepticism.

6. Design tutors that express uncertainty. Given evidence that LLMs rarely express uncertainty and are overconfident when they do (Zhou et al., 2024), educational AI should be tested on whether it communicates calibrated uncertainty. First-person uncertainty expressions reduced overreliance in a medical question-answering task (Kim et al., 2024); whether they work the same way for learners, and whether they affect motivation, is untested.

7. Treat introductions and policies as interventions. How instructors and institutions introduce an AI tool, and how grading rewards verification, plausibly shape students' trust (Pataranutaporn

et al., 2023; Vasconcelos et al., 2023). These low-cost levers should be evaluated with the same calibration measures as interface designs.

8. Study longer horizons and real courses. Trust has a learned layer alongside dispositional and situational layers (Hoff & Bashir, 2015). Semester-scale studies that track how students' calibration evolves as they encounter AI errors would address a central gap identified across the literature.

10. Conclusion

The evidence I reviewed supports a clear message: the educational problem is not that students trust generative AI too much or too little, but that their trust is often poorly calibrated, and that fluent explanations can make it worse. Interventions that communicate reliability can reduce overreliance on incorrect answers, but most were tested outside learning tasks, and none of the studies I reviewed measured learning without the AI. Interventions that require learners to think before accepting are the only family with direct evidence of preserved learning, yet they are less liked and tend to help the learners who were already best equipped. Generative AI makes the task harder, because the systems themselves rarely signal doubt. Calibrating learners' trust therefore requires designs that take the learning goal as seriously as the AI's accuracy, and research that asks, for every intervention, not only whether it works but for whom.

Declarations

Use of generative AI. I used Claude (Anthropic), a generative AI assistant, in preparing this manuscript. Claude searched bibliographic databases (Crossref, OpenAlex, PubMed, and arXiv), retrieved abstracts and open-access full texts, drafted and revised the text, built the reference list from Crossref metadata, and checked each cited claim against the source's published abstract

and, for the core studies, its full text. I chose the topic and scope, set the rule that only peer-reviewed and fully verified sources be cited, decided which sources and arguments to include, and reviewed and revised the manuscript. I take full responsibility for its content. No AI tool is listed as an author.

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